

CAST-RESIN DRY-TYPE DISTRIBUTION TRANSFORMERS

PDTR

Three-phase cast-resin dry-type distribution transformers for indoor installation
400 – 3150 kVA · 11 / 0.4 kV & 22 / 0.4 kV · Dyn11 · IEC 60076-11



3150 kVA

SINGLE UNIT UP TO

22 kV

PRIMARY VOLTAGE

F1 · E2 · C1

FIRE, ENVIRONMENT,
CLIMATE

IEC

60076-11 COMPLIANT

POWERLINE PDTR TRANSFORMERS

Since 2012 Powerline has grown from a trading company into a fully integrated engineering partner — an OEM for medium voltage switchgear and a certified ABB panel builder for low voltage boards, backed by two factories in 10th of Ramadan City and more than 2,200 delivered projects. The PDTR transformer completes the chain: MV switchgear, transformer and LV board from one supplier, engineered to work together.

PDTR at a glance

A three-phase cast-resin dry-type distribution transformer for indoor installation. The HV windings are vacuum-cast in epoxy resin — no oil, no fire load worth the name and no containment to build, so the transformer can stand inside the building next to the load it feeds.

- Eleven ratings, 400 to 3150 kVA, in one series
- Two primary voltages: 11 / 0.4 kV and 22 / 0.4 kV, both Dyn11 at 50 Hz
- Vacuum cast-resin HV windings, class F insulation throughout
- AN natural cooling, uprated with AF forced-air fans
- HV off-circuit tap changer, $+2 \times 2.5\%$ / $-4 \times 2.5\%$
- Fire class F1, environmental class E2, climatic class C1
- Supplied bare (IP00) or in an IP23 ventilated sheet-steel enclosure
- Designed, manufactured and tested to IEC 60076-11



PDTR CAST-RESIN DRY-TYPE TRANSFORMER

400 – 3150

KVA PER UNIT

11 & 22 kV

PRIMARY VOLTAGE

Dyn11

VECTOR GROUP

Class F

INSULATION SYSTEM

Model designation

PDTR · 11 · 40 · 102300 — series · primary voltage in kV (11 or 22) · rating index (40 = 400 kVA, 63 = 630, 31 = 3150) · series suffix for the 0.4 kV secondary, Dyn11, indoor, 50 Hz standard build. **PDTR1140102300** = 400 kVA, 11 / 0.4 kV.

Where it is used

Indoor substations in commercial and industrial buildings, basement and rooftop substations where oil is not permitted, hospitals, data centres, airports and metro, water and wastewater, cement, steel and chemicals, and compact substations built around the unit.

02 · TECHNOLOGY & FEATURES

TECHNOLOGY, FEATURES & CLASSES

The HV winding is wound, moulded and impregnated with epoxy resin under vacuum, then cured. What comes out is a solid, void-free cylinder in which the conductor, the insulation and the mechanical structure are one piece.



Vacuum casting

Casting under vacuum drives out air before the resin cures. Voids are where partial discharge starts, so removing them is what gives a cast-resin winding its dielectric life.



Fire class F1

The resin is self-extinguishing, with restricted smoke and halogen-free gas emission. There is no burning oil to feed a fire — the reason cast-resin units are specified for occupied buildings.



Environmental class E2

Rated for frequent condensation and heavy pollution. Cast surfaces can be cleaned and the unit returned to service; there is no conservator, breather or desiccant to maintain.

Safety

- Fire class F1 — self-extinguishing
- No oil, no bund, no fire wall
- Low smoke, halogen-free
- IP23 enclosure option

Performance

- Void-free vacuum-cast HV windings
- Class F insulation system
- Temperature rise < 100 K at 40 °C
- 2 s thermal short-circuit withstand

Environment

- Environmental class E2
- Climatic class C1
- Humidity up to 100 %
- Altitude up to 1,000 m

Installation

- Indoor, on rails or rollers
- No oil pit or containment
- Front access to terminals and taps
- AF uprating without changing the unit

C1

CLIMATIC CLASS

E2

ENVIRONMENTAL CLASS

F1

FIRE BEHAVIOUR CLASS

< 100 K

TEMPERATURE RISE AT 40 °C

No oil, no containment

An oil-filled unit of this size needs a bund, a fire wall or separation distance, and periodic oil sampling. A cast-resin unit needs none of it — the saving in civil works often exceeds the difference in transformer price.

Ventilation is still required

A dry-type transformer rejects its losses to the room air. Size the substation ventilation on the total loss $P_0 + P_k$ from the rating table so the air at the transformer stays within the 40 °C design ambient.

RATING TABLES — 11 & 22 KV

Design values at the 75 °C reference temperature. Dyn11, 50 Hz, AN cooling, class F insulation, indoor. Dimensions, weights, enclosure sizes and sound levels for the same ratings are on page 5.

11 / 0.4 kV — HV insulation level LI 75 kV / AC 35 kV, LV AC 5 kV

Model	Rating kVA	I _{HV} A	I _{LV} A	P ₀ W	P _k W	Total W	U _k %	η 100 % %
PDTR1140102300	400	21	577.4	1150	4300	5450	4.03	98.66
PDTR1150102300	500	26.2	721.7	1300	5100	6400	4.14	98.74
PDTR1163102300	630	33.1	909.3	1500	6400	7900	3.97	98.76
PDTR1180102300	800	42	1154.7	1700	7700	9400	4.92	98.84
PDTR1110102300	1000	52.5	1443.4	1850	8000	9850	5.1	99.02
PDTR1112102300	1250	65.6	1804.2	2500	10500	13000	5.11	98.97
PDTR1115102300	1500	78.7	2165.1	2600	11200	13800	5.85	99.09
PDTR1116102300	1600	84	2309.4	2800	12300	15100	5.85	99.07
PDTR1120102300	2000	105	2886.8	3500	14900	18400	5.97	99.09
PDTR1125102300	2500	131.2	3608.4	4300	18300	22600	6	99.10
PDTR1131102300	3150	165.3	4546.6	5500	22000	27500	7	99.13

22 / 0.4 kV — HV insulation level LI 125 kV / AC 55 kV, LV AC 5 kV

Model	Rating kVA	I _{HV} A	I _{LV} A	P ₀ W	P _k W	Total W	U _k %	η 100 % %
PDTR2240102300	400	10.5	577.4	1200	4800	6000	4.04	98.52
PDTR2250102300	500	13.1	721.7	1500	5600	7100	3.99	98.60
PDTR2263102300	630	16.5	909.3	1650	6800	8450	3.94	98.68
PDTR2280102300	800	21	1154.7	2000	8200	10200	5.09	98.74
PDTR2210102300	1000	26.2	1443.4	2200	8900	11100	5.02	98.90
PDTR2212102300	1250	32.8	1804.2	2800	11500	14300	4.91	98.87
PDTR2215102300	1500	39.4	2165.1	2800	12800	15600	5.99	98.97
PDTR2216102300	1600	42	2309.4	3100	14000	17100	5.97	98.94
PDTR2220102300	2000	52.5	2886.8	4000	17500	21500	6.02	98.94
PDTR2225102300	2500	65.6	3608.4	5000	20000	25000	5.92	99.01
PDTR2231102300	3150	82.7	4546.6	6300	23000	29300	6.94	99.08

P₀ no-load loss · P_k load loss · U_k impedance voltage, all at 75 °C. Efficiency is calculated at 100 % load and unity power factor from the declared losses: $\eta = S / (S + P_0 + P_k)$. Currents are at the rated tap. Tolerances to IEC 60076-1. U_k sets the LV short-circuit level and load sharing in parallel — order matched impedances where units are to run in parallel.

04 · DIMENSIONS

DIMENSIONS, WEIGHTS & ENCLOSURES

Transformer dimensions are length × width × height of the bare unit in millimetres. Enclosure dimensions are for the optional IP23 ventilated sheet-steel enclosure — all enclosures are 2,200 mm high. Sound level is the A-weighted sound pressure level at rated voltage.

11 / 0.4 kV

kVA	L × W × H	Weight kg	Enclosure	dB
400	1141×1120×1020	1240	1600×1450×2200	55
500	1191×1150×1020	1430	1650×1500×2200	56
630	1281×1230×1170	1735	1800×1650×2200	57
800	1121×1390×1170	2020	1900×1650×2200	58
1000	1336×1410×1120	2440	1900×1650×2200	59
1250	1371×1500×1120	2760	2000×1750×2200	61
1500	1412×1620×1270	3265	2200×1750×2200	61
1600	1391×1680×1320	3445	2200×1750×2200	61
2000	1516×1790×1370	4145	2200×1800×2200	62
2500	1626×1880×1370	4930	2400×1800×2200	65
3150	1641×2080×1470	5755	2600×1900×2200	65

22 / 0.4 kV

kVA	L × W × H	Weight kg	Enclosure	dB
400	1376×1180×1300	1495	1800×1800×2200	55
500	1506×1240×1300	1785	1800×1800×2200	56
630	1486×1300×1450	2060	1900×1950×2200	57
800	1411×1440×1450	2295	2000×1950×2200	58
1000	1491×1460×1500	2710	2000×2000×2200	59
1250	1641×1520×1500	3200	2100×2000×2200	61
1500	1616×1670×1550	3640	2250×2100×2200	61
1600	1616×1780×1600	3905	2300×2100×2201	61
2000	1716×1830×1650	4445	2400×2200×2200	62
2500	1846×1890×1650	5335	2400×2200×2200	65
3150	1841×2090×1750	6285	2700×2300×2200	65

Design values; certified outline and terminal drawings are issued with the order. Weights are for the bare transformer.

Planning the room

Allow clear access at the HV and LV terminal faces and space to roll the unit in. Where an IP23 enclosure is used, plan on the enclosure footprint plus door swing.

Handling

Lifting lugs on the top clamping frame, bi-directional rollers on the base. Lift with all four lugs and a spreader; never by the coils, terminals or enclosure.

Sound

55 dB at 400 kVA rising to 65 dB at 3150 kVA. Anti-vibration pads are available where the substation adjoins occupied space.

STANDARDS, CONDITIONS & TAP CHANGER

PDTR transformers are designed, manufactured and tested to the IEC 60076 series, with IEC 60076-11 as the governing standard for dry-type units.

Standards

IEC 60076-11	Dry-type transformers — the governing design, test and rating standard
IEC 60076-1	General requirements — ratings, tapplings, connections, tolerances
IEC 60076-3	Insulation levels, dielectric tests, external clearances
IEC 60076-5	Ability to withstand short circuit
IEC 60076-10	Determination of sound levels
IEC 60529	Degrees of protection provided by enclosures (IP code)

Insulation levels

Winding	11 kV	22 kV
HV lightning impulse (LI)	75 kV	125 kV
HV power frequency (AC)	35 kV	55 kV
LV power frequency (AC)	5 kV	5 kV

Service conditions

Service condition	Indoor
Cooling	AN / AF
Insulation class	F
Loss reference temperature	75 °C
Design ambient	40 °C
Temperature rise limit	< 100 K
Max altitude	1,000 m
Max humidity	100 %
Frequency	50 Hz
Vector group	Dyn11
Thermal short-circuit withstand	2 s
Transformer / enclosure / fan IP	IP00 / IP23 / IP ≥ 20

Above 1,000 m, or where the ambient exceeds 40 °C, the rating must be derated — contact Powerline engineering with the site conditions.

HV off-circuit tap changer (+2 × 2.5 % / -4 × 2.5 %)

Tap position	1	2	3	4	5	6	7
Step	+5 %	+2.5 %	rated	-2.5 %	-5 %	-7.5 %	-10 %
HV voltage · 11 kV series (V)	11550	11275	11000	10725	10450	10175	9900
HV voltage · 22 kV series (V)	23100	22550	22000	21450	20900	20350	19800

Position 3 is the rated tap; LV is 400 V on every tap. Taps are changed by moving the links on the HV coil face with the transformer isolated, proved dead and earthed — never under load. Move all three phases to the same position and record the new tap on the rating plate.

Vector group Dyn11

Delta on the high voltage side, star with a brought-out neutral on the low voltage side, LV lagging HV by 30°. The delta primary contains triplen harmonics; the star secondary provides the neutral the LV distribution system needs.

06 · OPTIONS, QUALITY & ORDERING

COOLING, OPTIONS QUALITY & ORDERING

PDTR is rated AN — natural air cooling — and can be uprated with forced-air fans without changing the transformer. Thermal protection is by sensors embedded in the LV windings reporting to a temperature relay.

Cooling & accessories

Item	Standard / option
AN natural cooling	Standard — the rating in the tables on page 4
AF forced-air fans	Option — IP ≥ 20, relay controlled
Winding temperature sensors	Standard — PT100 or PTC in each LV winding
Temperature relay	Option — alarm, trip, fan control, volt-free contacts
Rating plate	Standard, on the top frame
Earthing terminals	Standard, on the frame
Lifting lugs & rollers	Standard, bi-directional rollers
Off-circuit tap links	Standard, on the HV coil face
IP23 enclosure	Option, ventilated sheet steel
Anti-vibration pads	Option
Cable boxes / busbar flanges	Option, to project drawing

The AF uprating factor depends on the rating and fan arrangement and is confirmed by Powerline engineering with the order.

Information required when ordering

- Rated power in kVA
- HV and LV voltages and the tapping range
- Vector group, if other than Dyn11
- Impedance, where units run in parallel
- Cooling: AN only, or AN/AF with fans and relay
- Enclosure: bare IP00 or IP23
- Terminal arrangement — busbar flanges or cable boxes
- Site ambient, altitude and environmental conditions
- Monitoring and SCADA interface
- Witnessed test requirements

Note. Ratings and dimensions in this catalogue are design values for the standard build and are confirmed against the project specification at order stage.

Routine tests — every transformer

- Winding resistance, HV and LV
- Voltage ratio and phase displacement on every tap
- Short-circuit impedance and load loss
- No-load loss and no-load current
- Separate-source and induced AC withstand voltage tests
- Insulation resistance and partial discharge measurement

Type & special tests

- Temperature rise · lightning impulse withstand
- Sound level to IEC 60076-10
- C1 / E2 / F1 classification · short-circuit withstand

Documentation with every delivery

Routine test certificate, rating plate data, certified outline and terminal drawings, the unit data sheet, and the installation, operation and maintenance manual.



Performance excellence. Execution precision. Sustainable solutions.

From engineering and manufacture to testing, delivery and lifetime support — Powerline is your integrated partner for MV & LV power distribution in Egypt and the region.

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For technical enquiries or quotations, please get in touch with our team — we respond with engineered solutions, not just prices.

400 – 3150

KVA PER UNIT

11 & 22 kV

PRIMARY VOLTAGE

F1 · E2 · C1

FIRE / ENVIRONMENT /
CLIMATE

IEC 60076

-11 COMPLIANT